

Ang Kang Heng

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Summary

Strategic AI & Data Engineer with hands-on enterprise expertise in predictive analytics, multi-agent LLM orchestration, and high-speed data engineering. Proven ability to architect cost-optimized machine learning models (CatBoost, XGBoost) and multimodal extraction pipelines (VLMs, RAG) that successfully bypass hardware limits and commercial API constraints. Adept at transforming raw operational data into Explainable AI (SHAP) and automated dashboards, driving measurable efficiencies and equipping leadership with transparent, Day-1 preventative insights.

Experience

Zebra Technologies, CSI Specialist (Data, AI & Automation)

Penang, Malaysia

Apr 2026 – present

5 months

- Engineered a **highly concurrent Python API** with **SSE stream interception** to automate chatbot interactions. Processed 10,000-request workloads, accelerating data classification by **15x** (2 mins to 8 secs/row) and eliminating **300+ manual hours weekly**.
- Architected a **scalable multi-agent framework** using process synchronization for reliable agent execution. Integrated **Model Context Protocol (MCP)** and code-validation to enable secure, real-time tabular data manipulation.
- Engineered a **secure Python execution sandbox** for evaluating autonomous AI tasks. Built memory optimization and text-replacement tools, allowing agents to **autonomously audit and execute code modifications**, reducing manual review overhead by **>80%**.
- **Developed a Day-1 escalation predictive model** (LightGBM/CatBoost) achieving **>80% ROCAUC/F1**. Translated complex ML predictions into clear routing rules via **SHAP and Surrogate Trees**, deploying automated drift detection for preventative insights.
- **Diagnosed a core Google Gemini model regression** affecting enterprise operations. Led a 1.5-month escalation, leveraging performance evidence to definitively prove a vendor deployment failure and drive **direct remediation by Google's Product Engineering team**.

Zebra Technologies, IT Intern (AI & Automation)

Penang, Malaysia

Oct 2025 – Apr 2026

7 months

- **Engineered an 8-agent hierarchical AI prototype (LangGraph, ReAct)** automating enterprise data synthesis. Integrated **Reflective Memory (ReMe)** and an **LLM-as-Judge** validation layer to enforce self-correction and rigorously verify AI reasoning.
- **Reduced web testing cycles from days to 90 minutes** deploying a modular Web Assurance Suite (**Python, Playwright**). Automated validation for 2.9M+ legacy HTML links and ensured 100% routing coverage across 114 modern **SPA** endpoints.
- **Engineered a Human-in-the-Loop (HITL) ETL pipeline (Python, Pandas)** bypassing strict API constraints. Utilized **LanceDB** semantic embeddings and **JSON prompt injection** to automate data sanitization and dynamic charting.
- **Optimized multilingual feedback triage** by deploying **Chain-of-Thought (CoT)** prompt engineering within **Power Automate**. Increased CES categorization accuracy to **98.25%** and entirely eliminated manual data entry.

Education

B.IT Universiti Teknikal Malaysia Melaka (UTeM), Bachelor of Information Technology (Games Technology) with Honours Oct 2022 – Apr 2026
CGPA: 3.93/4

Achievements

- **Dean's List Award (UTeM)**: Maintained a 3.93/4.00 CGPA by achieving a GPA above 3.50 for **all** semesters.
- **Gold Award & Industry Award (GDIID 2025)**: Swept three major categories for the architectural design of Astraeus Protocol, recognized by industry judges for technical excellence, algorithmic innovation, and commercial viability.
- **1st Place (Best Game) & 3rd Place (Best Poster)**, Workshop II Final Showcase: Outperformed 10+ teams in technical execution, innovation, and project communication for the development of 'Moral Nexus'.

Projects

AI Act Compliance Suite (Governance) | Python, Gemini API, SQLite Jan 2026 – Feb 2026

- **Designed 5 modular Agent Plugins** decoupling specialized AI agents to achieve **autonomous, zero-touch compliance verification** for 3 EU legislative articles.
- **Engineered a documentation engine** parsing 20+ file types across 500k-character contexts, automating the generation of **complex visual architectures (Mermaid.js)**.
- **Standardized cross-team system integration** by deploying a High Availability fallback framework, ensuring operational resilience and **guaranteeing 99.9% system uptime**.

Skills

Agentic AI: Multi-Agent Swarm Orchestration, LangGraph, State Machines, LLMs & VLMs, ReAct/CoT Logic, Prompt/Context Engineering, RAG, OCR, Vector Embeddings (LanceDB)

Machine Learning & Predictive Analytics: Ensemble Models (CatBoost, XGBoost, LightGBM, Random Forest), Explainable AI / XAI (SHAP, Surrogate Trees), NLP (TF-IDF, Topic Modeling)

Data Engineering & Architecture: Python (Polars, Pandas), Rust-based Parsers (Calamine), Power BI (DAX, Semantic Modeling), ETL Pipelines, Parquet, Data Sanitization

Process & Web Automation: Power Automate, Server-Sent Events (SSE), Playwright (Shadow DOM, Context Pooling), Async Python, CI/CD (GitHub Actions)

Programming Languages: Python, JavaScript, SQL, HTML/CSS

Languages: Mandarin, English, Malay, Hokkien